SAMANTHA (SAM) LUI

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EDUCATION

Bachelor of Computer Science (Third Year) | University of British Columbia *September 2024 – May 2027*
Advanced Diploma – Clinical Genetics (88%) | British Columbia Institute of Technology *August 2020 – October 2021*
Bachelor of Health Sciences (Dean's Honour Roll) | Simon Fraser University *September 2011 – December 2017*

PROFESSIONAL EXPERIENCE

CIHI – Applied Cloud Architecture Developer Intern | North York, O.N. *January 2026 – Present*

- Migrating 3 enterprise **Java** monolithic applications from Spring to **Spring Boot** by refactoring 20+ **Maven** dependencies, modernizing deprecated APIs and standardizing configurations, improving cloud deployment stability onto **Amazon EKS**
- Resolved containerization failures related to **JAR** packaging and **JSP** frameworks by transitioning to **Thymeleaf** while optimizing production-ready **Dockerfiles** to enable automated image creation with **Bamboo** and **Docker** for **Amazon ECR**
- Authoring and maintaining 5+ internal **Confluence** documentation pages outlining migration and troubleshooting workflows, improving team onboarding efficiency and knowledge transfer while reducing repeated debugging efforts

Shared Services Canada – I.T. Analyst Co-op Student | Vancouver, B.C. *August 2025 – December 2025*

- Delivered **Windows 11** imaging, and managed **Microsoft Azure** user states with **MDM** and **SCCM**, leveraging **PowerShell** automation for secure account provisioning and system performance optimization
- Developed an internal automation tool with **Python**, **Tkinter** and **Selenium**, streamlining workflows with reduced manual data entry errors and Excel template auto-population, improving ticket completion speed by **75%**
- Authored a **60+** page training guide standardizing support procedures, adopted as a core onboarding resource for new co-ops across regional support teams

Provincial Health Services Authority – Clinical Genetics Technologist | Vancouver, B.C *July 2021 – August 2024*

- Executed high-throughput genetic analyses under standardized protocols, ensuring timely and accurate results reporting
- Liaised with IT/vendors to **troubleshoot** LIS and equipment software/hardware issues, reducing operational downtime

TECHNICAL PROJECTS

OverHacked! | *Phaser.js, Next.js, FastAPI, LangChain* | GitHub *January 2026 – January 2026*

- Built **FastAPI** backend endpoints to support agentic AI workflows, integrating OpenAI LLMs with **CrewAI** to dynamically generate topic-specific programming quizzes for mini-game interactions
- Co-developed AI agents using **LangChain**, enabling personalized question generation based on player experience level and preferred programming language

AniMori – Anime Library App | *Electron.js, React, Vite, SQLite, Node.js, Jikan* | GitHub *August 2025 – December 2025*

- Developed a cross-platform **Electron.js** + **React** desktop app with automated metadata retrieval via the **Jikan API**
- Implemented **IPC** channels for client-to-backend connectivity, and optimized **SQLite** database schema with **DBeaver** to streamline data flow and improve query efficiency

JobQuest AI – AI-Powered Job App Assistant | *Next.js, AWS Bedrock, ElevenLabs* | GitHub *October 2025 – October 2025*

- Designed a gamified job search assistant with a **Next.js** video-game style UI, increasing engagement via interactive resume reviews and voice-based mock behavioural interviews
- Planned and prototyped AI-powered features (**AWS Bedrock**, **ElevenLabs**), delivering personalized, score-based insights

SipLy – Beverage Intake Visualization Tool | *React, Spring Boot, H2, Spring AI, Groq* | GitHub *July 2025 – Present*

- Scaling a Java application into a **React** + **Spring Boot** web app with **Recharts** for dynamic beverage intake visualizations
- Designed **RESTful CRUD APIs** with **H2** and secure **BCrypt** authentication, ensuring reliable data storage
- Planning integration of **Spring AI** with **Groq LLaMA3** for personalized beverage recommendations, based off user data

TECHNICAL SKILLS

Languages | Python, Java, C++, PowerShell (currently acquiring), SQL, R, HTML5, CSS3, JavaScript
Frameworks/Libraries | React.js, Node.js/Express.js, Next.js, Electron.js, Spring Boot, JUnit Jupiter, FastAPI
Databases | PostgreSQL, MongoDB (NoSQL), SQLite, H2
Cloud & Containerization Platforms | Docker, JFrog Artifactory, Kubernetes, Amazon Web Services (ECR, EKS)
Developer Tools | Git/GitHub, GitLab, Atlassian (Confluence, Bamboo, Jira), Netlify, Postman, Vite, Jupyter, Groq, Selenium
Enterprise Systems | Microsoft Azure (AD, Mobile Device Management, Microsoft Endpoint Configuration Manager)